Yicheng Yang

Curriculum Vitae

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Research interests

Arithmetic geometry and the Langlands programme, p-adic and mod p modular forms, Galois representations - specifically: mod p coherent cohomology of Shimura varieties, higher Hida theory, local-global compatibility

Education

2022–present **Ph.D. in Mathematics**, Imperial College London. Supervisor: George Boxer and Fred Diamond, Expected completion: June 2027.

2017-2021 **Bachelor of Science in Mathematics**, Fudan University.

Publications and preprints

1. **Weight one modular forms of level $\Gamma_1(p)$ and local-global compatibility**, link, 2026.

Research talks

Sep 2026 *Local-global compatibility of weight one modular forms.* Young researchers in Algebraic Number Theory(Y-RANT 2026), University of Bristol.

Sep 2025 *Integral weight one forms.* Young researchers in Algebraic Number Theory(Y-RANT 2025), University of Nottingham.

Nov 2024 *Weight one modular forms of level $\Gamma_0(p)$ in characteristic p.* London Junior Number Theory Seminar, Kings College London.

Seminar and Study Groups Organised

Apr 2026-Jun 2026 **Study group on the Fargues-Fontaine curve**, with Simon Alonso, Calle Sonne, and Haoran Liang.

Oct 2025-Apr 2026 **Study group on higher Coleman theory**, with Haoran Liang.

Selected Conferences Attended

Jun 2026 Categorical Langlands Summit, KIAS, Seoul, South Korea

Jun 2026 Classical and p-adic aspects of the Kudla program, CIRM, Luminy, France

Jan 2026 Around the Langlands program, CIRM, Luminy, France

Oct 2025 New Structures and Techniques in p-adic Geometry, IHES, Bures-sur-Yvette, France

Oct 2025 Non-Archimedean geometry, Lille, France

Aug 2025	Geometric Aspects of the Langlands Correspondence, Essen, Germany
Oct 2024	Workshop on p-adic geometry, National University of Singapore, Singapore
Sep 2024	New Advances in the Langlands Program: Geometry and Arithmetic, Oxford, UK
Jul 2024	p-adic Families of Automorphic Forms: Theories and Applications, ICMS, Edinburgh, UK
Jun 2024	Modular Forms, L-functions, and Eigenvarieties: a conference in memoriam of Joël Bellaïche, ENS, Paris, France
Sep 2023	Arithmétique des formes automorphes, LMO, Orsay, France

Awards

2022-2027	**EPSRC studentship**

Teaching experience

Spring 2026	**Graduate Teaching Assistant**, Imperial College London. Linear Algebra and Groups
Fall 2025	**Graduate Teaching Assistant**, Imperial College London. Groups and Rings